

Student Number:

Maths Class: _____



James Ruse Agricultural High School

2020

HSC Task 1

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 1 hour and 20 minutes
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- Show relevant mathematical reasoning and/or calculations
- Answers to the Multiple Choice questions and Question 7(a) should be completed on the separate sheet and handed in with the rest of question 7.

Total Marks:
70

Section I – 5 marks

- Attempt Questions 1–5

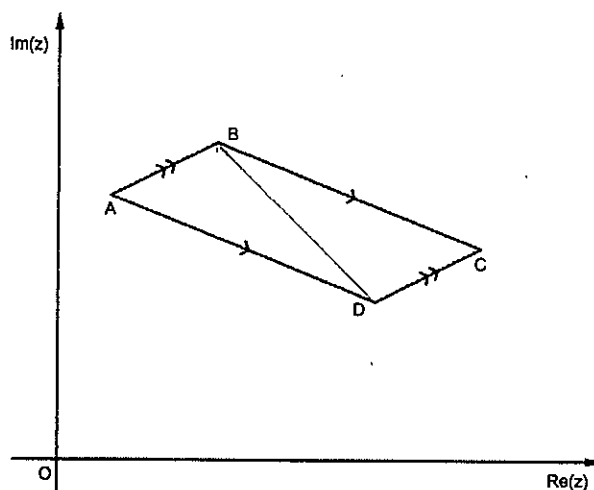
Section II – 65 marks

- Attempt Questions 6–10

Section 1 – Multiple Choice (5 Marks)

Attempt questions 1 – 5

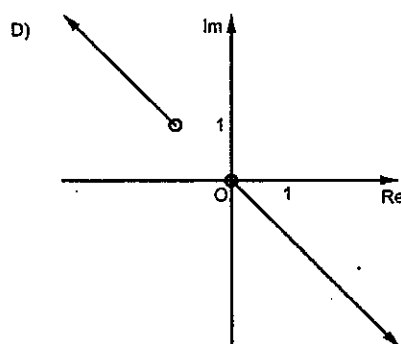
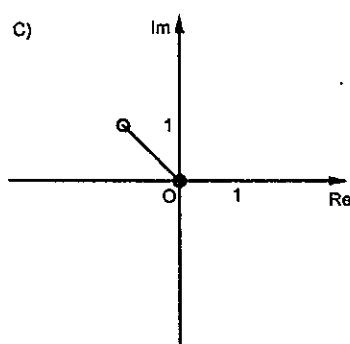
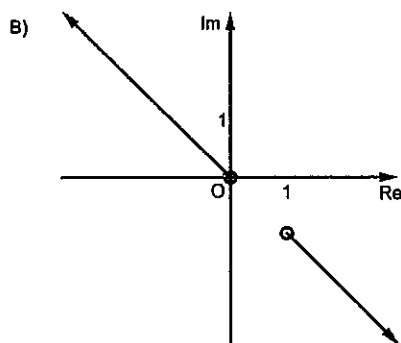
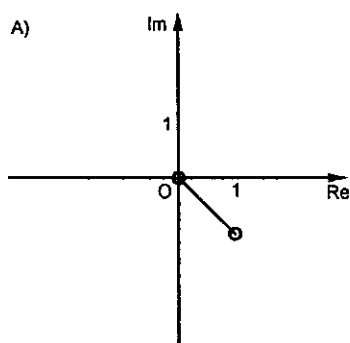
Record your answers on the tear-off sheet provided.



1. The above diagram shows a parallelogram $ABCD$ drawn in the first quadrant of the complex plane. The points A , B and C represent the complex numbers z_1 , z_2 and z_3 respectively. The vector $\overrightarrow{DB}$ represents the complex number
 - A) $z_1 + z_3 - 2z_2$
 - B) $z_2 - z_1 - z_3$
 - C) $2z_2 - z_1 - z_3$
 - D) $2z_2 - z_1 + z_3$
2. Which of the following is equal to i^i ?
 - A) -1
 - B) 1
 - C) $e^{-\frac{\pi}{2}}$
 - D) $e^{\frac{\pi}{2}}$
3. If $z_1 = -2i$ and $z_2 = 1 - i\sqrt{3}$, which one of the following statements is true?
 - A) $\arg(z_1 + z_2) = -\frac{5\pi}{12}$
 - B) $|z_1 - z_2| = 2 + \sqrt{3}$
 - C) $\arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{6}$
 - D) $\arg(z_1 z_2) = -\frac{\pi}{3}$

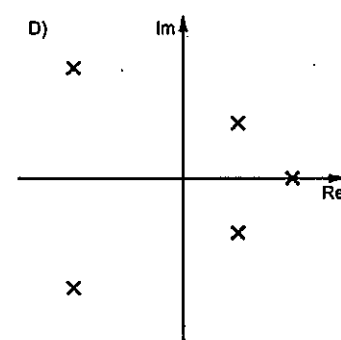
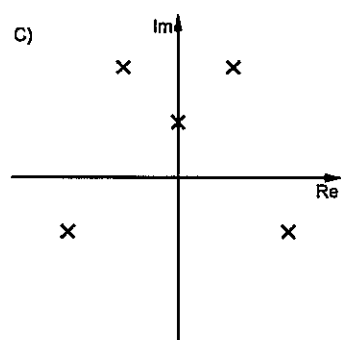
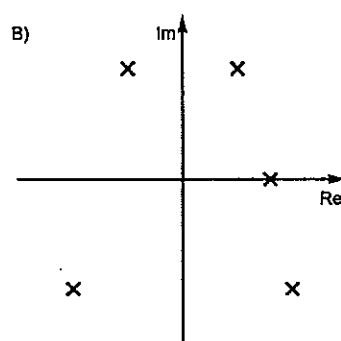
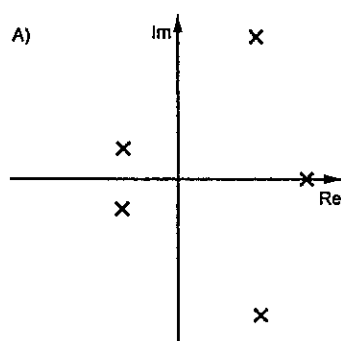
4. Which of the following diagrams best represents the solutions to the equation

$$\arg(z) = \arg(z + 1 - i)$$



5. Which of the following Argand planes could represent the position of the roots of

$$z^5 + z^2 - z + k = 0 \quad \text{where } k \text{ is a real number?}$$



Section 2 – (65 Marks)

Start each of questions 6 – 10 on a fresh piece of paper.
Each of these questions is worth 13 marks.

Question 6. (13 Marks) Start on a fresh sheet of paper

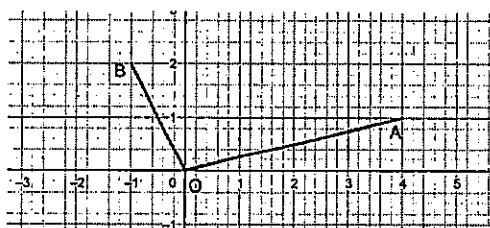
Marks

- a) Let $z = 1 + 2i$ and $w = 3 - i$.
Find, in the form $x + iy$ (x, y real),
- i) $z\bar{w}$ ii) $\overline{\left(\frac{10}{z}\right)}$ 2
- b) Let $\alpha = 1 - i$
- i) Write α in mod-arg form and exponential form. 2
- ii) Hence show that $\alpha^4 + 4 = 0$. 1
- c) Show that the complex number $z = \frac{6-2i}{3+4i} - \frac{6}{5i}$ is real. 2
- d) i) Solve the equation $z^6 + 8 = 0$, writing the roots in exponential form. 3
- ii) Illustrate these roots on a clear Argand diagram. 1
- iii) Write $z^6 + 8$ in factorised form over the field of real numbers. 2

Question 7. (13 Marks) Start on a fresh sheet of paper

Marks

- a) On the Argand diagram below the points A and B represent the complex numbers α and β respectively.



On the tear-off sheet provided construct the points P, Q, R, S and T which respectively represent :

- i) $\alpha + \beta$ ii) $\bar{\alpha}$ iii) $-\beta$ iv) $\beta - \alpha$ v) $2i\beta$ 5

Ensure that the diagram sheet (including Multiple Choice answers) is attached to the other answers of question 7.

(Question 7 continues on the next page.)

Question 7 (continued)**Marks**

- b) The complex number z satisfies $|z - 2 + 4i| = 3$ and w satisfies $\arg(w + 2 + 3i) = \frac{\pi}{4}$. The point P represents z on an Argand diagram and Q represents w .
- i) Sketch the loci of P and Q on a clear Argand diagram. 4
- ii) If $w = x + iy$ (x, y being real numbers), write down the equation of the locus of Q in terms of x and y . 2
- iii) Hence deduce the minimum value of $|z - w|$ in surd form. 2

Question 8. (13 Marks) Start on a fresh sheet of paper**Marks**

- a) i) Find all real numbers x and y that satisfy $(x + iy)^2 = 20i - 21$. 3
- ii) Hence solve the equation $z^2 + 6z + 30 - 20i = 0$. 2
- b) Find all the complex numbers $z = a + ib$, where a and b are real, such that $|z|^2 + 5\bar{z} + 10i = 0$. 3
- c) You may assume for this question that, if $z = \cos\theta + i\sin\theta$ then $\left(z^n + \frac{1}{z^n}\right) = 2 \cos n\theta$ for positive integers n . Let $w = z + \frac{1}{z}$.
- i) Prove that $w^3 + w^2 - 2w - 2 = \left(z + \frac{1}{z}\right) + \left(z^2 + \frac{1}{z^2}\right) + \left(z^3 + \frac{1}{z^3}\right)$. 2
- ii) Hence, or otherwise, find all solutions of $\cos\theta + \cos 2\theta + \cos 3\theta = 0$ in the range $0 \leq \theta \leq 2\pi$. 3

Question 9. (13 Marks) Start on a fresh sheet of paper**Marks**

- a) i) If $z = \cos\theta + i\sin\theta$, use de Moivre's Theorem to prove that

$$\left(z^n - \frac{1}{z^n}\right) = 2i \sin n\theta . \quad 2$$

- ii) Hence, or otherwise, prove that $16\sin^5\theta = \sin 5\theta - 5\sin 3\theta + 10\sin\theta$ 3

- b) On an Argand diagram shade and label the region specified by 3

$$\{z : \operatorname{Im}(z) \leq 4\} \cap \{z : |z - 4 - 5i| \leq 3\}$$

- c) The points P and Q represent the complex numbers z and w respectively on an Argand diagram where $w = \frac{3z+2}{z-1}$. 3

By writing an expression for z in terms of w , or otherwise, determine and sketch the locus of Q as P traces out the circle $|z| = 1$.

- d) If $\operatorname{Arg}\left(\frac{z_1+z_2}{z_1-z_2}\right) = \frac{\pi}{2}$ show that $|z_1| = |z_2|$. 2

Question 10. (13 Marks) Start on a fresh sheet of paper**Marks**

- a) i) Show that, if $w = \cos\frac{2\pi}{7} + i\sin\frac{2\pi}{7}$, then the equation $z^7 - 1 = 0$ has distinct roots $1, w, w^2, w^3, w^4, w^5$ and w^6 . 2

- ii) Explain why the equation $z^6 + z^5 + z^4 + z^3 + z^2 + z + 1 = 0$ has roots w, w^2, w^3, w^4, w^5 and w^6 . 1

- iii) Find the value of $\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7}$. 2

- iv) Find the monic quadratic equation with numerical coefficients whose roots are $w + w^2 + w^4$ and $w^3 + w^5 + w^6$. 2

- b) The number c is real and non-zero. It is also known that $(1 + ic)^5$ is real.

- i) Expand $(1 + ic)^5$ and deduce that $c^4 - 10c^2 + 5 = 0$. 2

- ii) Hence show that $c = \sqrt{5 - 2\sqrt{5}}, -\sqrt{5 - 2\sqrt{5}}, \sqrt{5 + 2\sqrt{5}}$ or $-\sqrt{5 + 2\sqrt{5}}$. 1

- iii) Let $1 + ic = re^{i\theta}$. Show that the smallest possible positive value of θ is $\frac{\pi}{5}$. 1

- iv) Hence determine the exact value of $\tan\frac{\pi}{5}$, justifying your answer. 2

Q6

2020 7-11/12 4th Q6

V2

a) $z = 1 + 2i$ $w = 3 - i$

i) $z\bar{w} = (1 + 2i)(3 + i)$

$$= 3 + 7i - 2$$

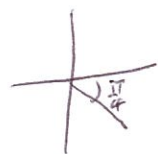
$$= 1 + 7i$$

ii) $\left(\frac{10}{z}\right) = ?$ $\frac{10}{1+2i} \times \frac{1-2i}{1-2i} = \frac{10-20i}{5}$ 1 m

$\left(\frac{10}{z}\right) = \frac{10+20i}{5} = 2+4i$ 1 m

b) i) $\alpha = 1 - i = \sqrt{2} \left[\cos\left(\frac{\pi}{4}\right) + i \left(\sin\left(-\frac{\pi}{4}\right)\right) \right]$ 1 m

$$= \sqrt{2} e^{-\frac{\pi}{4}i}$$
 1 m



ii) $\alpha^4 = \left(\sqrt{2} e^{-\frac{\pi}{4}i}\right)^4 = (\sqrt{2})^4 \left(\cos\left(-\frac{\pi}{4} \times 4\right) + i \sin\left(-\frac{\pi}{4} \times 4\right)\right)$ 1 m
(D. Moivre's Th)
 $= 4 \times \cos(-\pi)$
 $= 4 \times -1 = -4$ 1 m

$$\therefore \alpha^4 + 4 = -4 + 4 = 0$$
 1 m

c) $z = \frac{6-2i}{3+4i} - \frac{6}{5i}$

$$\frac{6-2i}{(3+4i)(3-4i)} = \frac{18-8-6i-24i}{9+16} = \frac{10-30i}{25} = \frac{2-6i}{5}$$
 1 m

$$\frac{6}{5i} \times \frac{5i}{5i} = \frac{30i}{-25} = -\frac{6i}{5}$$

$$\therefore z = \frac{2-6i}{5} - \frac{6i}{5} = \frac{2}{5} \in \mathbb{R}$$
 1 m

$$b) i) z^6 = -8$$

$$= 2^3 e^{(\pi + 2k\pi)i}, k \in \mathbb{Z}$$

$$z = (\sqrt{2}) e^{\frac{(\pi + 2k\pi)}{6}i}$$

1m for $\sqrt{2}$

1m for $(\pi + 2k\pi)/6$

1m for final answer

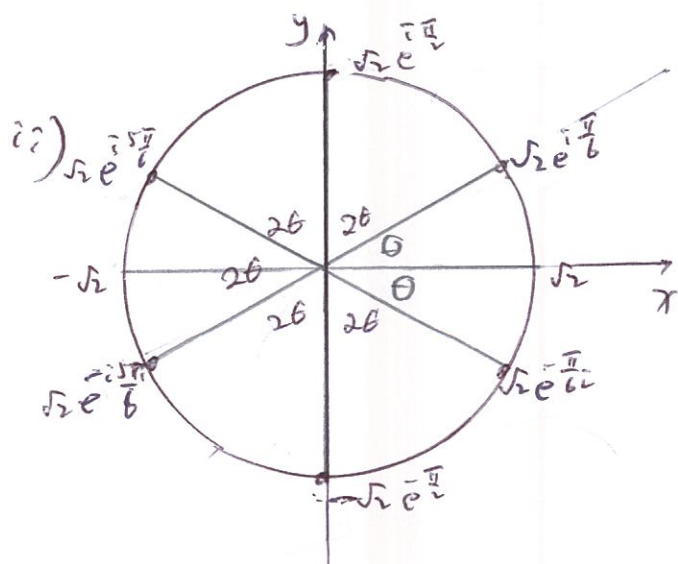
Roots are $z = \sqrt{2} e^{-i(\frac{\pi}{6})}, \sqrt{2} e^{i\frac{\pi}{6}}, \sqrt{2} e^{-i(\frac{\pi}{2})}, \sqrt{2} e^{i\frac{\pi}{2}}, \sqrt{2} e^{-i\frac{5\pi}{6}}, \sqrt{2} e^{i\frac{5\pi}{6}}$

Alternatively $z = r \cos \theta$
 $z^6 = r^6 \cos 6\theta$ (D.M. 74)

$$r^6 \cos(6\theta) = 2^3 \cos(\pi + 2k\pi)$$

$$r = \sqrt{2}, \quad \theta = \frac{\pi + 2k\pi}{6}$$

$\therefore$ Roots are $\sqrt{2} e^{i\frac{\pi}{6}}, \sqrt{2} e^{-i\frac{\pi}{6}}, \sqrt{2} e^{i\frac{\pi}{2}}, \sqrt{2} e^{-i\frac{\pi}{2}}, \sqrt{2} e^{i\frac{5\pi}{6}}, \sqrt{2} e^{-i\frac{5\pi}{6}}$



$$\theta = \frac{\pi}{6}$$

must show $r = \sqrt{2}$ 1m
 and $\theta = \frac{\pi}{6}$

• Angles need to be evenly distributed around circle

$$iii) z^6 + 8 = (z^2)^3 + 2^3 = (z^2 + 2)(z^4 - 2z^2 + 4)$$

$$= (z^2 + 2)[(z^4 + 4z^2 + 4) - 6z^2]$$

$$= (z^2 + 2)(z^2 + 2 - \sqrt{6}z)(z^2 + 2 + \sqrt{6}z)$$

Alternatively

$$z^6 + 8 = (z - \sqrt{2} e^{i\frac{\pi}{2}})(z - \sqrt{2} e^{-i\frac{\pi}{2}})(z - \sqrt{2} e^{i\frac{\pi}{6}})(z - \sqrt{2} e^{-i\frac{\pi}{6}})(z - \sqrt{2} e^{i\frac{5\pi}{6}})(z - \sqrt{2} e^{-i\frac{5\pi}{6}})$$

$$z^6 + 8 = (z^2 - \sqrt{2}i)(z^2 + \sqrt{2}i)(z^2 - 2\sqrt{2} \cos \frac{\pi}{6} z + 2)(z^2 - 2\sqrt{2} \cos \frac{5\pi}{6} z + 2)$$

Since $(z - z_1)(z - \bar{z}_1) = z^2 - 2(\operatorname{Re} z_1)z + |z_1|^2$

$$= (z^2 + 2)(z^2 - 2\sqrt{2} \frac{\sqrt{3}}{2} z + 2)(z^2 - 2\sqrt{2} \frac{-\sqrt{3}}{2} z + 2)$$

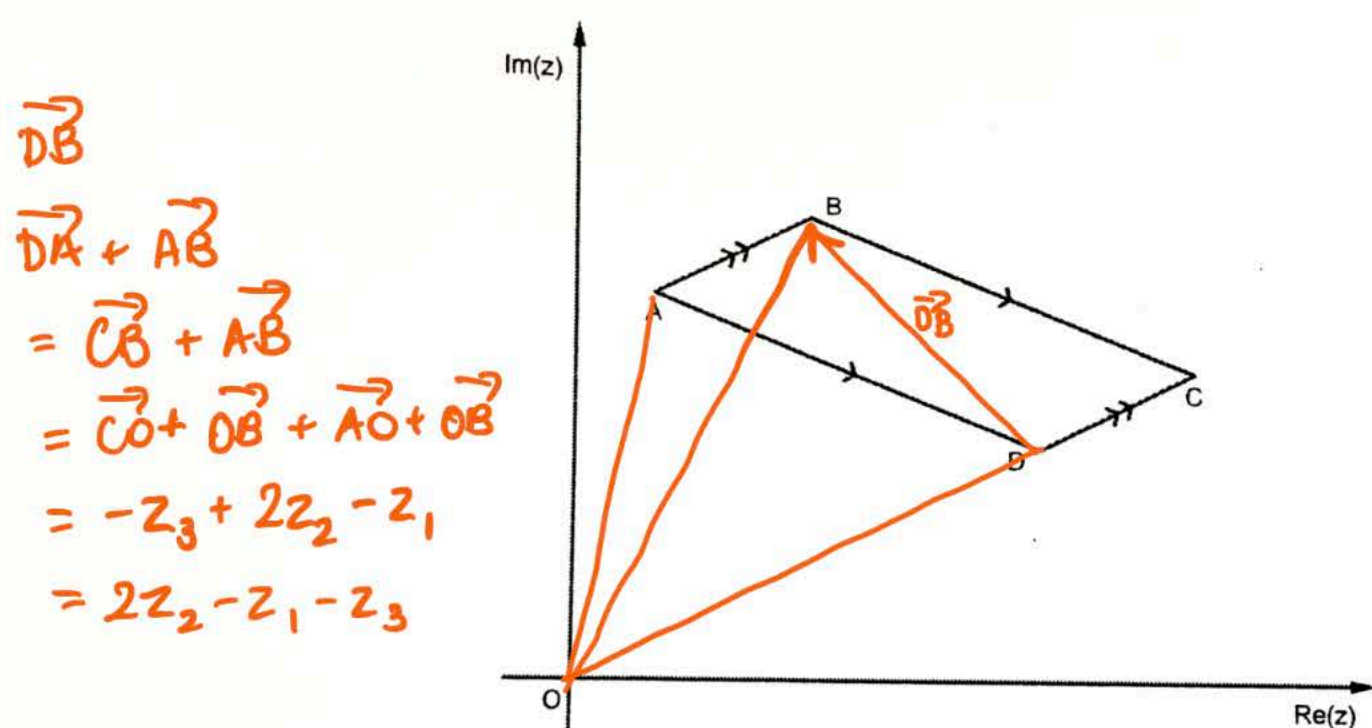
$$= (z^2 + 2)(z^2 - \sqrt{6}z + 2)(z^2 + \sqrt{6}z + 2)$$

★ many students forget z with $\sqrt{6}$, $|z|^2 \neq 1$

Multiple Choice solutions

Monday, 23 November 2020 2:43 pm

1.



1. The above diagram shows a parallelogram $ABCD$ drawn in the first quadrant of the complex plane. The points A, B and C represent the complex numbers z_1, z_2 and z_3 respectively. The vector $\vec{DB}$ represents the complex number

A) $z_1 + z_3 - 2z_2$

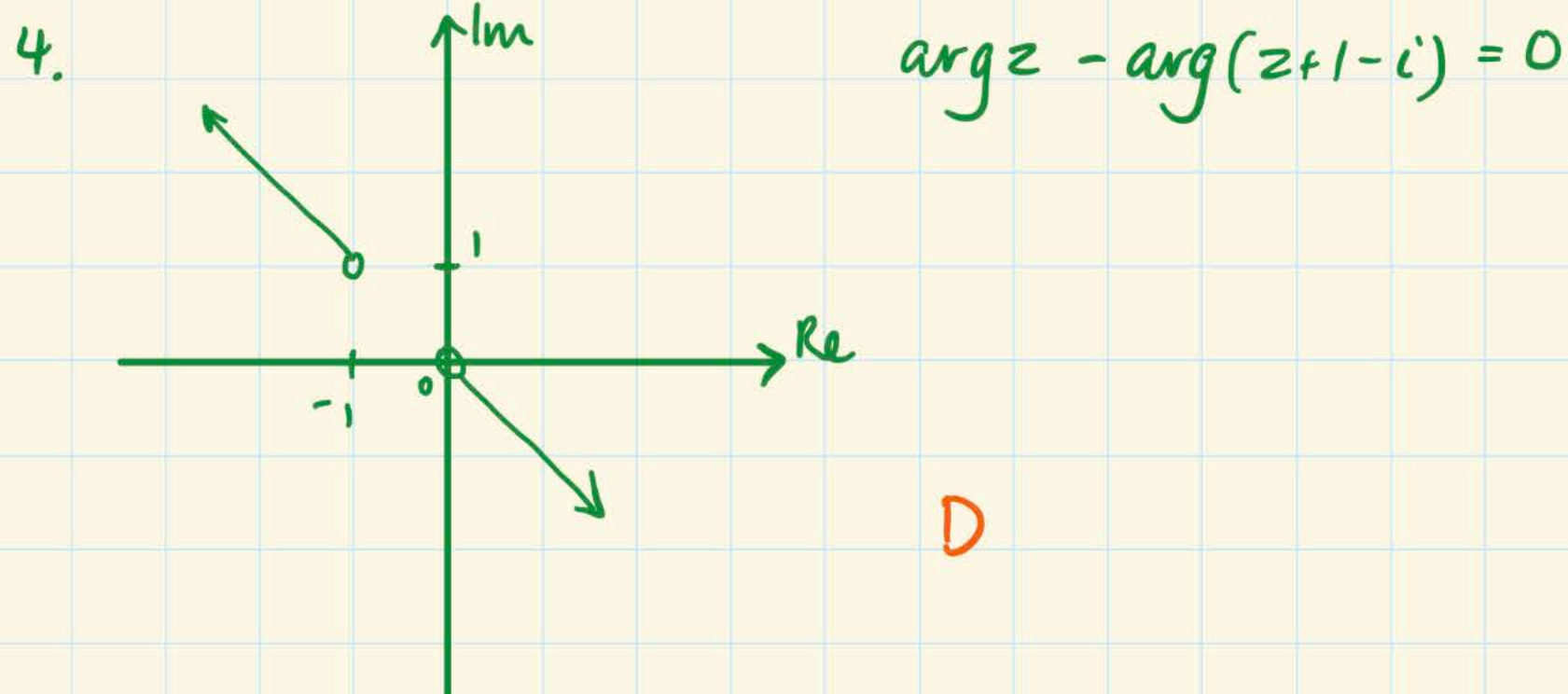
B) $z_2 - z_1 - z_3$

☒ C) $2z_2 - z_1 - z_3$

D) $2z_2 - z_1 + z_3$

2. $i^i = (e^{i\pi/2})^i = e^{-\pi/2}$ C

3. $z_1 + z_2 = 1 - (\sqrt{3} + 2)i$
 $\arg(z_1 + z_2) = \tan^{-1} -(\sqrt{3} + 2)$
 $= -1.31$ calculator
 $= -5\pi/2$ A



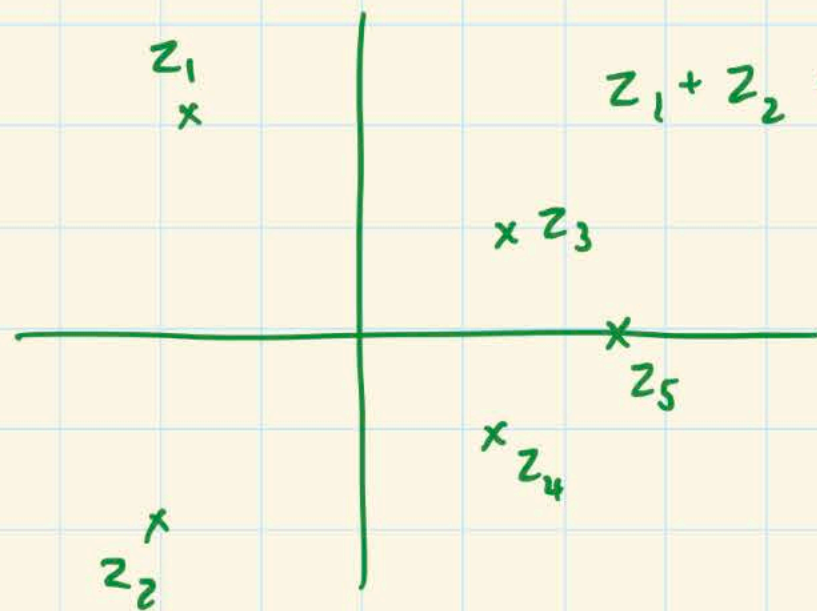
5. Real coefficients

$\therefore$ Complex roots occur in conjugate pairs

$\therefore$ A or D

For z & $\bar{z}$, $z + \bar{z} = \text{Re}(z)$

$\therefore$ Sum of roots is real
& sum of roots = 0



$$z_1 + z_2 < 0, \quad z_3 + z_4 + z_5 > 0$$

$$\therefore z_1 + z_2 = -(z_3 + z_4 + z_5)$$

only possible when
 $\text{Re}(z_3)$ is small.

$\therefore$ D

Student number _____

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Multiple Choice Answers

Circle the letter of the most correct answer.

1. A B **C** D

2. A B **C** D

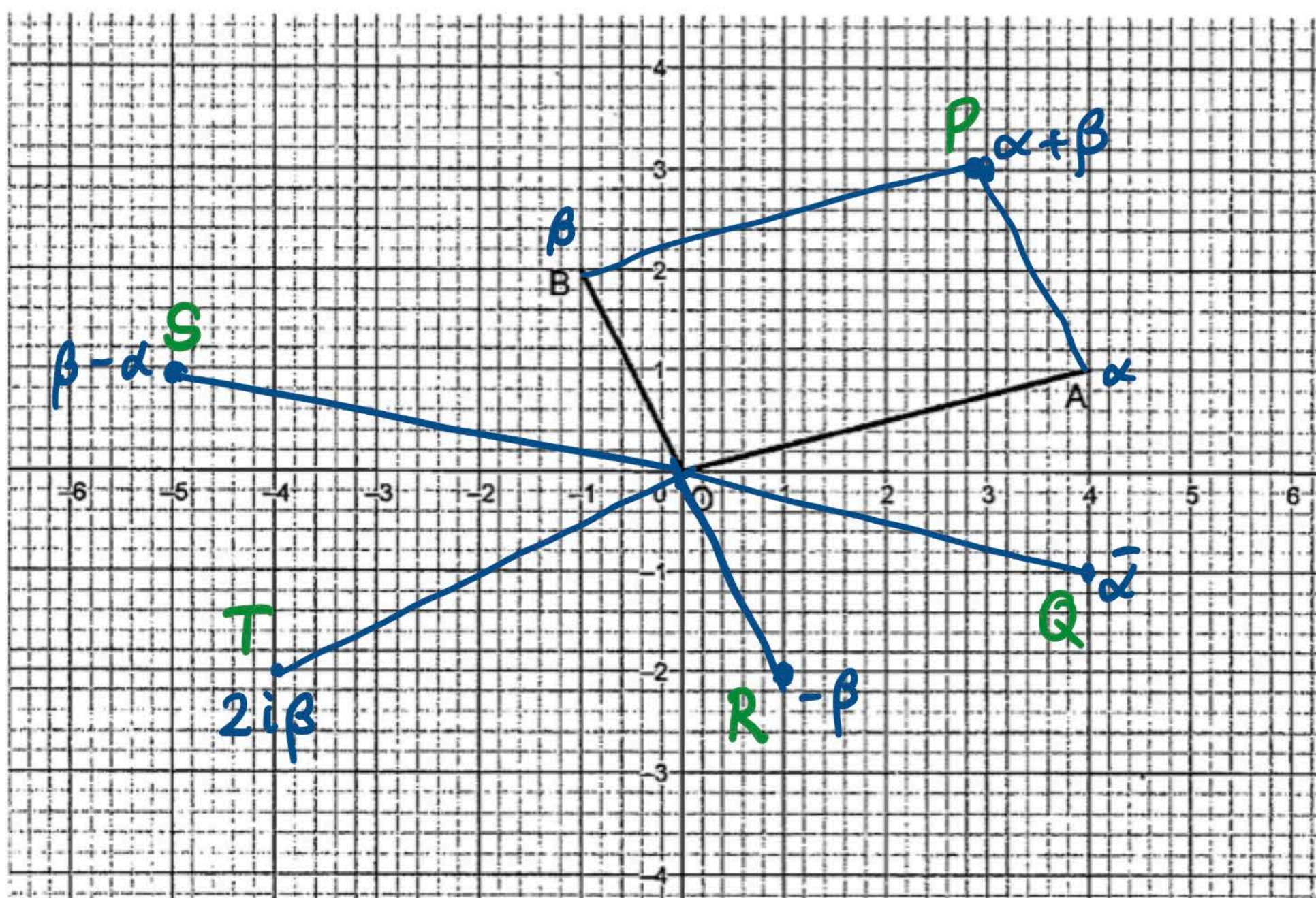
3. **A** B C D

4. A B C **D**

5. A B C **D**

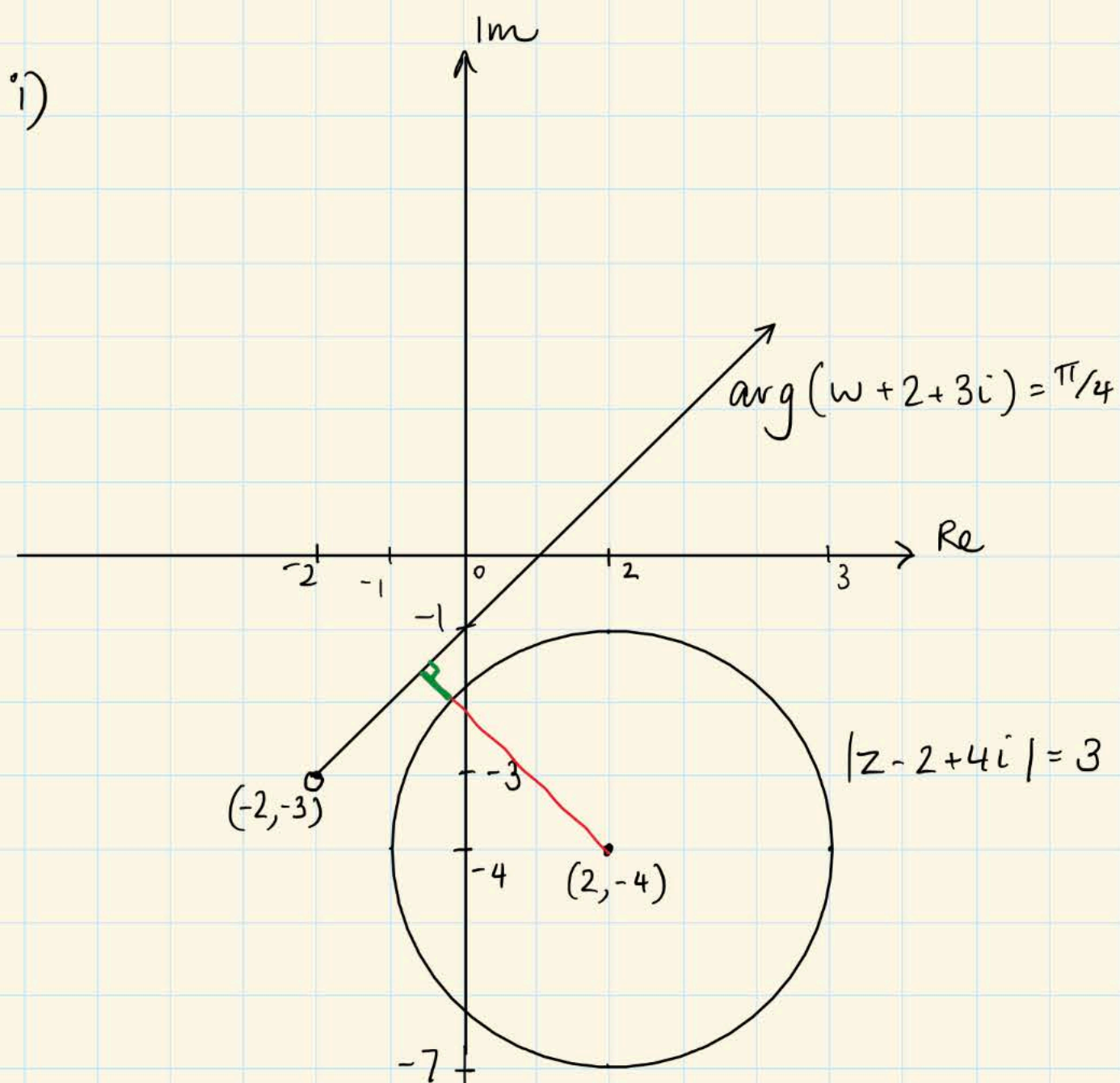
MC Mark: /5

Question 7a) Your answers to question 7(a) should be drawn and labelled on this diagram.



This sheet should be submitted as the top sheet of question 7.

i)



$$|z - 2 + 4i| = 3$$

Circle, centre (2, -4)

radius 3

- 1m circle centre (2, -4)
- 1m radius 3 shown

$$\arg(w + 2 + 3i) = \frac{\pi}{4}$$

- 1m open circle (-2, -3)
- 1m straight line
- 1m indicate $\pi/4$
 - pass thru (0, -1)
 - $\pi/4$ labelled.

(ii) $m = 1, b = -1$
 $y = x - 1 \Leftrightarrow x - y - 1 = 0$

- 1m gradient of 1
- 1m equation

(iii) minimum distance, perpendicular distance from line to circle

$$\text{Centre to line} = \frac{|1 \times 2 - 1 \times -4 - 1|}{\sqrt{1^2 + 1^2}}$$

$$= \frac{5}{\sqrt{2}}$$

$$\text{minimum value of } |z - w| = \frac{5}{\sqrt{2}} - 3$$

$$= \frac{5\sqrt{2} - 6}{2}$$

- 1m perp distance
Centre (2, -4) to
line $x - y - 1 = 0$

$$\text{1m min } |z - w| = \frac{5\sqrt{2} - 6}{2} \text{ or } \frac{5}{\sqrt{2}} - 3.$$

$$\text{or } \boxed{3}$$

OR tangent parallel to $x-y-1=0$
is of the form $x-y+k=0$

distance from tangent to centre is 3.

$$(2, -4), \quad x-y+k=0$$

$$\therefore \frac{|2 - (-4) + k|}{\sqrt{2}} = 3$$

$$|6+k| = 3\sqrt{2}$$

$$k = -6 + 3\sqrt{2} \quad \text{or} \quad k = -6 - 3\sqrt{2}$$

too big

$$\therefore k = -6 + 3\sqrt{2}$$

$$\therefore \text{Tangent is } x-y-6+3\sqrt{2}=0$$

$(0, -1)$ is a point on $x-y-1=0$

perpendicular distance from

$$(0, -1) \text{ to } x-y-6+3\sqrt{2}=0$$

$$d = \frac{|1 \times 0 - 1 \times (-1) - 6 + 3\sqrt{2}|}{\sqrt{2}}$$

$$= \frac{|-5 + 3\sqrt{2}|}{\sqrt{2}}$$

$$= \frac{5 - 3\sqrt{2}}{\sqrt{2}} < \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{5\sqrt{2} - 6}{2}$$

• 1m tangent

• 1m perp distance

MATHEMATICS Extension 2: Question...8...

Suggested Solutions	Marks	Marker's Comments
a) i) $(x+iy)^2 = 20i - 21$ $x^2 - y^2 + 2xyi = 20i - 21$ equating real and imaginary parts $x^2 - y^2 = -21$ - ① $2xy = 20$ $xy = 10$ $y = \frac{10}{x}$ - ② sub. ② into ① $x^2 - \left(\frac{10}{x}\right)^2 = -21$ $x^4 + 21x^2 - 100 = 0$ $(x^2 - 4)(x^2 + 25) = 0$ $x^2 = 4$, $x \in \mathbb{R}$ $\therefore x = 2$, $y = 5$ $x = -2$, $y = -5$ $\therefore x + iy = \pm(2 + 5i)$	1 1 1	getting x or y and eliminating the imaginary values matching the correct x and y values together.
<u>OR</u> $x^2 - y^2 = -21$ - ① $2xy = 20$ $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ $= (-21)^2 + (20)^2$ $= 841$ $\therefore x^2 + y^2 = 29$ - ② ① + ② : $2x^2 = 8$ $x^2 = 4$ $x = \pm 2$ $\therefore x = 2$, $y = 5$ $x = -2$, $y = -5$	1 1	getting x or y correct and having $x^2 + y^2 = 29$
ii) $z = \frac{-6 \pm \sqrt{36 - 4(30 - 20i)}}{2}$ $= \frac{-6 \pm \sqrt{80i - 84}}{2}$ $= -3 \pm \sqrt{20i - 21}$	1	many students had transcription error and wrote the equation as: $z^2 - 6z + 30 - 20i = 0$ no marks were deducted.

MATHEMATICS Extension 2: Question 8

Suggested Solutions	Marks	Marker's Comments
$= -3 + (2 + 5i) \quad \text{or} \quad -3 - (2 + 5i)$ $= -1 + 5i \quad \text{or} \quad -5 - 5i$	1	
b) $z ^2 + 5\bar{z} + 10i = 0$ $a^2 + b^2 + 5(a - ib) + 10i = 0$ $a^2 + b^2 + 5a - 5bi + 10i = 0$ equating real and imaginary parts $a^2 + b^2 + 5a = 0 \quad \text{--- ①}$ $-5b + 10 = 0$ $5b = 10$ $b = 2$ sub. $b = 2$ into ① $a^2 + 4 + 5a = 0$ $(a + 4)(a + 1) = 0$ $\therefore a = -1, -4$ $\therefore z = -1 + 2i \quad \text{or} \quad -4 + 2i$	1	
c) i) LHS = $w^3 + w^2 - 2w - 2$ $= \left(z + \frac{1}{z}\right)^3 + \left(z + \frac{1}{z}\right)^2 - 2\left(z + \frac{1}{z}\right) - 2$ $= z^3 + 3z + 3\left(\frac{1}{z}\right) + \frac{1}{z^3} + z^2 + 2 + \frac{1}{z^2}$ $- 2z - 2\left(\frac{1}{z}\right) - 2$ $= z^3 + \frac{1}{z^3} + z^2 + \frac{1}{z^2} + z + \frac{1}{z}$ $= \left(z + \frac{1}{z}\right) + \left(z^2 + \frac{1}{z^2}\right) + \left(z^3 + \frac{1}{z^3}\right)$ $= \text{RHS}$	1	for substituting correctly
ii) $\cos \theta + \cos 2\theta + \cos 3\theta = 0$ $2\cos \theta + 2\cos 2\theta + 2\cos 3\theta = 0 \quad \downarrow \text{multiply by 2}$ $z + \frac{1}{z} + z^2 + \frac{1}{z^2} + z^3 + \frac{1}{z^3} = 0$ $w^3 + w^2 - 2w - 2 = 0$ $(w + 1)(w^2 - 2) = 0$ $\therefore w = -1, \pm\sqrt{2}$ $\therefore 2\cos \theta = -1, \quad 2\cos \theta = \pm\sqrt{2}$ $\cos \theta = -\frac{1}{2}, \quad \cos \theta = \pm\frac{\sqrt{2}}{2}$ $\theta = \frac{2\pi}{3}, \frac{4\pi}{3} \quad \theta = \frac{\pi}{4}, \frac{7\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$	1	explaining that the solution for $\cos \theta + \cos 2\theta + \cos 3\theta = 0$ is the same as the solutions for $w^3 + w^2 - 2w - 2 = 0$

MATHEMATICS Extension 2: Question...8...

Suggested Solutions

Marks

Marker's Comments

OR $\cos \theta + \cos 2\theta + \cos 3\theta = 0$

$$\cos \theta + 2\cos^2 \theta - 1 + \cos \theta \cos 2\theta - \sin \theta \sin 2\theta = 0$$

$$\cos \theta + 2\cos^2 \theta - 1 + \cos \theta (2\cos^2 \theta - 1)$$

$$- 2\sin^2 \theta \cos \theta = 0$$

$$\cos \theta + 2\cos^2 \theta - 1 + 2\cos^3 \theta - \cos \theta$$

$$- 2(1 - \cos^2 \theta) \cos \theta = 0$$

$$2\cos^2 \theta - 1 + 2\cos^3 \theta - 2\cos \theta + 2\cos^3 \theta = 0$$

$$4\cos^3 \theta + 2\cos^2 \theta - 2\cos \theta - 1 = 0$$

$$2\cos^2 \theta (2\cos \theta + 1) - (2\cos \theta + 1) = 0$$

$$(2\cos^2 \theta - 1)(2\cos \theta + 1) = 0$$

$$\therefore \cos \theta = \pm \frac{1}{\sqrt{2}} \quad \cos \theta = -\frac{1}{2}$$

$$\theta = \frac{\pi}{4}, \frac{7\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4} \quad \theta = \frac{2\pi}{3}, \frac{4\pi}{3}$$

1

1

1

MATHEMATICS Extension 2: Question 9

Suggested Solutions

Marks

Marker's Comments

a) i let $z = \cos\theta + i\sin\theta$

$$z^n = \cos n\theta + i\sin n\theta \quad (\text{De Moivre's})$$

$$\frac{1}{z^n} = z^{-n} = \cos(-n\theta) + i\sin(-n\theta) \quad (\text{De Moivre's})$$

$$= \cos n\theta - i\sin n\theta$$

$$\therefore z^n - \frac{1}{z^n} = (\cos n\theta + i\sin n\theta) - (\cos n\theta - i\sin n\theta)$$

$$= 2i\sin n\theta$$

ii (Hence)

$$2i\sin\theta = z - \frac{1}{z}$$

$$(2i)^5 \sin^5\theta = \left(z - \frac{1}{z}\right)^5$$

$$= z^5 - 5z^3 + 10z - 10z^{-1} + 5z^{-3} - z^{-5}$$

$$= (z^5 - z^{-5}) - 5(z^3 - z^{-3}) + 10(z - z^{-1})$$

$$= (2i\sin 5\theta) + 5(2i\sin 3\theta) + 10(2i\sin\theta) \quad \text{From i)}$$

$$32i\sin^5\theta = 2i\sin 5\theta + 10i\sin 3\theta + 20i\sin\theta$$

$$16\sin^5\theta = \sin 5\theta + 5\sin 3\theta + 10\sin\theta$$

(1) Use of De Moivre's

(1)

Using identity from i)

(1)

(1)

(1)

MATHEMATICS Extension 2: Question.....

Suggested Solutions

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ii (otherwise)

$$(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta \quad (\text{De Moivre's Theorem})$$

$$(\cos \theta + i \sin \theta)^5 = \cos^5 \theta + 5i \cos^4 \theta \sin \theta - 10 \cos^3 \theta \sin^2 \theta - 10i \cos^2 \theta \sin^3 \theta + 5 \cos \theta \sin^4 \theta + i \sin^5 \theta$$

Equating imaginary parts

$$\begin{aligned} \sin 5\theta &= 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta \\ &= 5(1 - \sin^2 \theta)^2 \sin \theta - 10(1 - \sin^2 \theta) \sin^3 \theta + \sin^5 \theta \\ &= 5 \sin \theta - 10 \sin^3 \theta + 5 \sin^5 \theta - 10 \sin^3 \theta + 10 \sin^5 \theta + \sin^5 \theta \\ &= 5 \sin \theta - 20 \sin^3 \theta + 16 \sin^5 \theta \end{aligned}$$

$$(\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta \quad (\text{De Moivre's Theorem})$$

$$(\cos \theta + i \sin \theta)^3 = \cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta$$

Equating Imaginary parts:

$$\begin{aligned} \sin 3\theta &= 3 \cos^2 \theta \sin \theta - \sin^3 \theta \\ &= 3(1 - \sin^2 \theta) \sin \theta - \sin^3 \theta \\ &= 3 \sin \theta - 3 \sin^3 \theta - \sin^3 \theta \\ &= 3 \sin \theta - 4 \sin^3 \theta \end{aligned}$$

Sub $\sin 3\theta$ and $\sin 5\theta$ into RHS:

$$\begin{aligned} \text{RHS} &= (5 \sin \theta - 20 \sin^3 \theta + 16 \sin^5 \theta) - 5(3 \sin \theta - 4 \sin^3 \theta) + 10 \sin \theta \\ &= 5 \sin \theta - 20 \sin^3 \theta + 20 \sin^3 \theta - 15 \sin \theta + 20 \sin^3 \theta + 10 \sin \theta \\ &= 16 \sin^5 \theta \\ &= \text{LHS} \end{aligned}$$

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Expression for $\sin 5\theta$

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Expression for $\sin 3\theta$

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b) let $z = x + iy$

$$\operatorname{Im}(z) \leq 4 \Rightarrow y \leq 4$$

$$|z - 4 - 5i| \leq 3$$

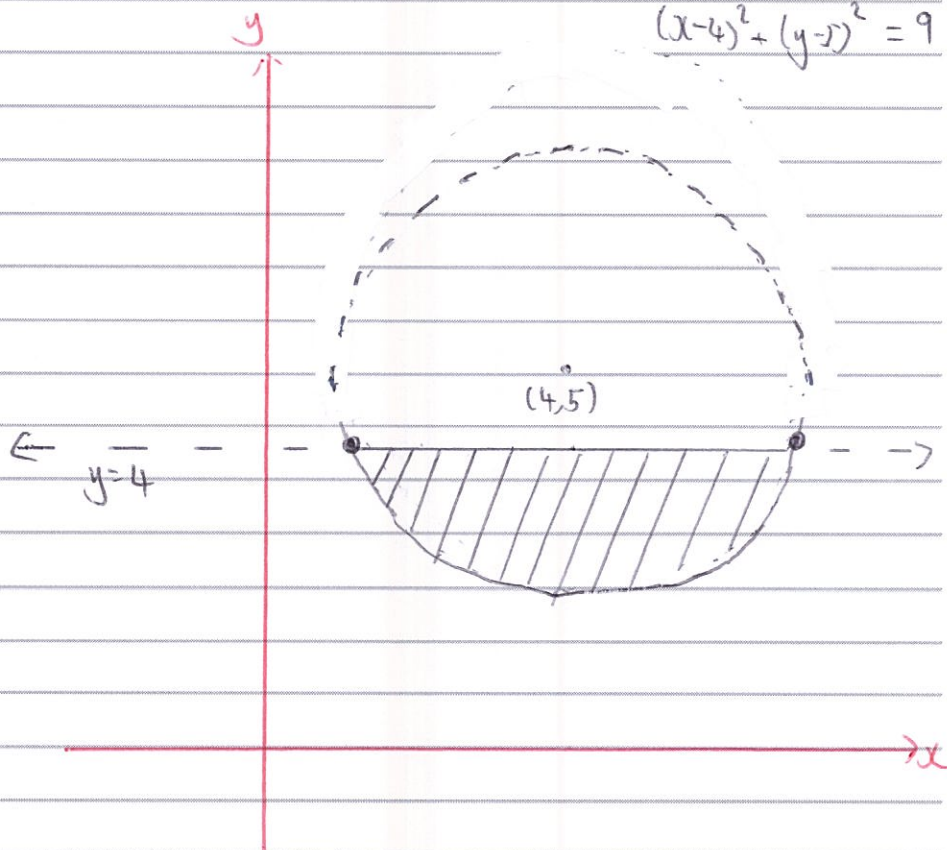
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$$|(x-4) + (y-5)i| \leq 3$$

$$\sqrt{(x-4)^2 + (y-5)^2} \leq 3$$

$$(x-4)^2 + (y-5)^2 \leq 9$$

$$(x-4)^2 + (y-5)^2 = 9$$



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$$c) w = \frac{3z + 2}{z - 1}$$

$$wz - w = 3z + 2$$

$$wz - 3z = w + 2$$

$$z(w - 3) = w + 2$$

$$z = \frac{w + 2}{w - 3}$$

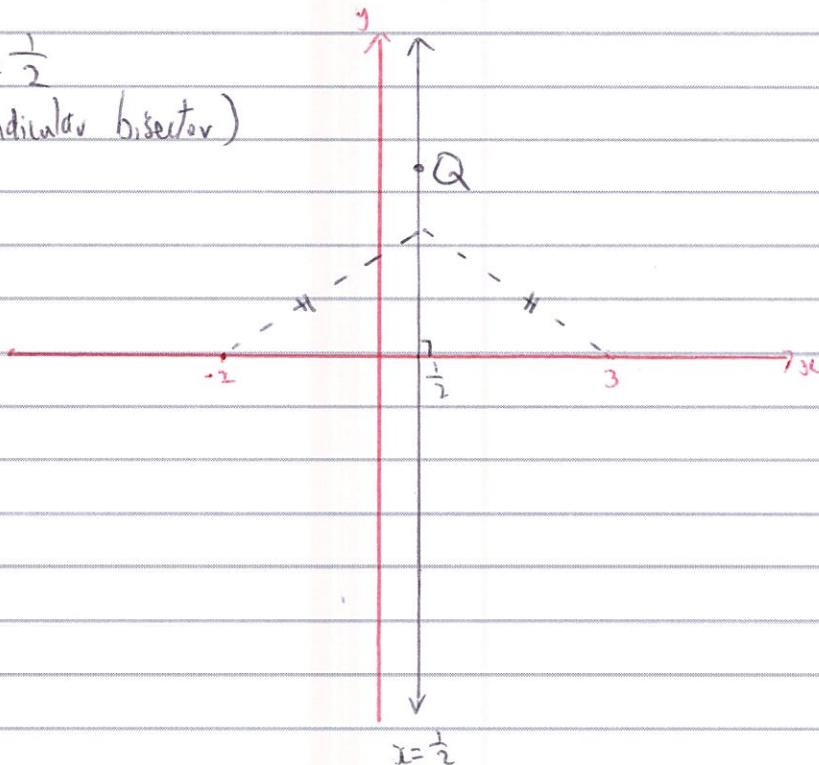
$$|z| = 1 \Rightarrow \left| \frac{w + 2}{w - 3} \right| = 1$$

$$\frac{|w + 2|}{|w - 3|} = 1$$

$$|w + 2| = |w - 3|$$

$$\therefore x = \frac{1}{2}$$

(perpendicular bisector)



MATHEMATICS Extension 2: Question.....

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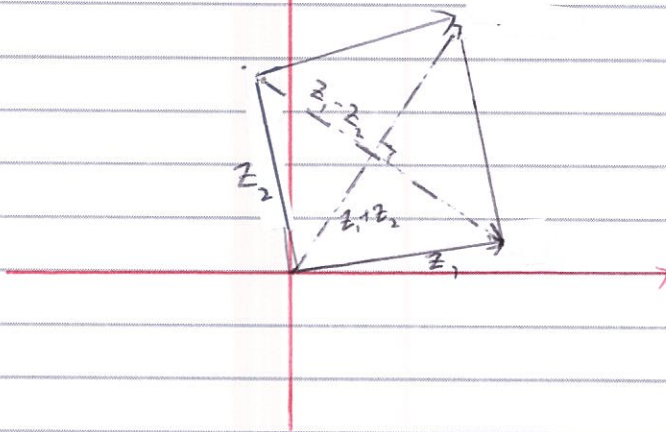
Marker's Comments

$$d) \operatorname{Arg} \left(\frac{z_1 + z_2}{z_1 - z_2} \right) = \frac{\pi}{2}$$

$$\operatorname{Arg}(z_1 + z_2) - \operatorname{Arg}(z_1 - z_2) = \frac{\pi}{2}$$

$$\operatorname{Arg}(z_1 + z_2) = \operatorname{Arg}(z_1 - z_2) + \frac{\pi}{2}$$

$$\therefore \underline{z_1 + z_2} \perp \underline{z_1 - z_2}$$



Since $\underline{z_1 + z_2}$ and $\underline{z_1 - z_2}$ are diagonals of the parallelogram formed by the origin, and points representing z_1 , z_2 , and $z_1 + z_2$ respectively, if $\underline{z_1 + z_2} \perp \underline{z_1 - z_2}$, then the parallelogram is a rhombus (diagonals are perpendicular).

$$\therefore |z_1| = |z_2|, \text{ (adjacent sides of a rhombus.)}$$

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d) Alternate Method.

$$\operatorname{Arg} \left(\frac{z_1 + z_2}{z_1 - z_2} \right) = \frac{\pi}{2}$$



$$\frac{z_1 + z_2}{z_1 - z_2} = ki \quad (k \in \mathbb{Z})$$

$$z_1 + z_2 = ki(z_1 - z_2)$$

$$z_1 + z_2 = kiz_1 - kiz_2$$

$$z_2 + kiz_2 = kiz_1 - z_1$$

$$z_2(1 + ki) = z_1(-1 + ki)$$

$$|z_2(1 + ki)| = |z_1(-1 + ki)|$$

$$|z_2| |1 + ki| = |z_1| |-1 + ki|$$

$$\text{Since } |1 + ki| = |-1 + ki| = \sqrt{1 + k^2}$$

$$\therefore |z_1| = |z_2|$$

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MATHEMATICS Extension 2: Question...10

Suggested Solutions	Marks	Marker's Comments
10. (a) (i) By Subs $z = 1$ is a solution Sub $z = \cos \frac{2n\pi}{7} + i \sin \frac{2n\pi}{7}, n \in \mathbb{Z}$ into equation $\begin{aligned} \text{LHS} &= \left(\cos \frac{2n\pi}{7}\right)^7 - 1 \\ &= \cos 2n\pi - 1 \quad (\text{de Moivre}) \\ &= \cos 2n\pi + i \sin 2n\pi - 1 \\ &= 1 - 1 \\ &= 0 \\ &= \text{RHS} \end{aligned}$ $\therefore w^n$ is a solution for integer n $1, w, w^2, \dots, w^6$ have distinct arguments, so the 7 roots are distinct, other values of n will be duplicates of one of these. (ii) $z^7 - 1 = 0 \therefore (z-1)(1+z+z^2+z^3+z^4+z^5+z^6) = 0$ $\therefore z = 1$ or $1+z+z^2+z^3+z^4+z^5+z^6 = 0$ Since the roots of $z^7 - 1 = 0$ are $1, w, w^2, w^3, w^4, w^5$ and w^6 $\therefore$ the roots of $1+z+z^2+z^3+z^4+z^5+z^6 = 0$ are w, w^2, w^3, w^4, w^5 and w^6. (iii) $1+w+w^2+w^3+w^4+w^5+w^6 = 0$ $w^3 \left(\frac{1}{w^3} + \frac{w}{w^3} + \frac{w^2}{w^3} + \frac{w^3}{w^3} + \frac{w^4}{w^3} + \frac{w^5}{w^3} + \frac{w^6}{w^3} \right) = 0, w \neq 0$ $\therefore w^3 + \frac{1}{w^3} + w^2 + \frac{1}{w^2} + w + \frac{1}{w} + 1 = 0$ $w^n + w^{-n} = 2 \cos n\theta$ $\therefore 2 \cos 3\theta + 2 \cos 2\theta + 2 \cos \theta + 1 = 0$ $2 \cos \frac{6\pi}{7} + 2 \cos \frac{4\pi}{7} + 2 \cos \frac{2\pi}{7} = -1$ $\therefore \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}$	1 1 1 1	

MATHEMATICS Extension 2: Question.....		
Suggested Solutions	Marks	Marker's Comments
(iv) let $\alpha = w + w^2 + w^4$ and $\beta = w^3 + w^5 + w^6$ $\alpha + \beta = w + w^2 + w^3 + w^4 + w^5 + w^6 = -1$ $\alpha\beta = (w + w^2 + w^4)(w^3 + w^5 + w^6)$ $= w^4(1 + w + w^3)(1 + w^2 + w^3)$ $= w^4(1 + w + w^5 + w^3 + w^4 + w^5 + w^6 + 2w^3)$ $= w^4(0 + 2w^3) = 2w^7 = 2$ The monic quadratic equation is $z^2 + z + 2 = 0.$	1	
(b) (i) $(1+ic)^5 = 1 + 5(ic) + 10(ic)^2 + 10(ic)^3 + 5(ic)^4 + (ic)^5$ $= 1 + 5ic - 10c^2 - 10ic^3 + 5c^4 + ic^5$ $= 1 - 10c^2 + 5c^4 + i(5c - 10c^3 + c^5)$ $(1+ic)^5$ is real $\therefore 5c - 10c^3 + c^5 = 0$ Since $c \neq 0 \therefore c^4 - 10c^2 + 5 = 0$	1	
(ii) $c^4 - 10c^2 + 5 = 0$ $c^4 - 10c^2 + 25 = 20$ $(c^2 - 5)^2 = 20$ $c^2 - 5 = \pm 2\sqrt{5}$ $c^2 = 5 \pm 2\sqrt{5}$ $c = \sqrt{5-2\sqrt{5}}, \sqrt{5+2\sqrt{5}}, -\sqrt{5-2\sqrt{5}}, \text{ or } -\sqrt{5+2\sqrt{5}}$	1	
(iii) $1+ic = re^{i\theta} \therefore (1+ic)^5 = r^5 e^{i5\theta}$ is real $\therefore 5\theta = 0, \pi, \dots$ The smallest positive value is $\pi \therefore \theta = \frac{\pi}{5}$	1	
(iv) $1+ic = re^{i\pi/5} = r(\cos \frac{\pi}{5} + i \sin \frac{\pi}{5})$ $r \cos \frac{\pi}{5} = 1$ and $r \sin \frac{\pi}{5} = c \therefore \tan \frac{\pi}{5} = c$ $0 \leq \frac{\pi}{5} \leq \frac{\pi}{2}$ in the first quadrant $\tan \frac{\pi}{5} > 0 \therefore \tan \frac{\pi}{5} = \sqrt{5-2\sqrt{5}}.$	1	